

ENGR. HUZAIFA SHAHBAZ

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PROFESSIONAL SUMMARY

Computer Engineering graduate with extensive hands-on experience in embedded systems, IoT development, PCB design, and software engineering. Proven expertise in ESP32, STM32, Arduino microcontrollers, digital system design, machine learning, and full-stack development. Demonstrated ability to deliver industrial-grade solutions including smart monitoring systems with cloud integration, AI-powered medical diagnosis systems, and complex network architectures.

TECHNICAL SKILLS

Programming: C/C++, Python, Java, Verilog HDL, JavaScript, MATLAB **Embedded Systems:** ESP32-S3, STM32, Arduino, 8051, LoRa/LoRaWAN, UART, PWM, GPIO **Tools & Platforms:** Xilinx Vivado, Android Studio, Visual Studio Code, Proteus, MySQL, Firebase, Flutter **Design & CAD:** PCB Design, SolidWorks, AutoCAD, MS Visio, Figma **Networking:** Cisco Packet Tracer, OSPF, VLAN, VPN, TCP/IP, Network Security

WORK EXPERIENCE

R&D Trainee | Machadev Engineering Pvt Ltd | Sep 2025 - Nov 2025

- Designed industrial-grade smart water monitoring and purification device using ESP32-S3 with 5 integrated sensors

Intern | Machadev Engineering Pvt Ltd | Jul 2025 - Aug 2025

- Gained experience in embedded systems design, PCB simulation, testing workflows, and SolidWorks CAD design

Intern | Risetech, NUST EME College | Jul 2024 - Sep 2024

- Developed expertise in ESP32-S3, STM32Cube, 8051 microcontrollers, LoRa/LoRaWAN, and PCB design workshops

EDUCATION

Bachelor of Science in Computer Engineering | HITEC University, Rawalpindi | Sep 2021 - Aug 2025

FSc (ICS) in Science General | Army Public School and College EME, Rawalpindi | Nov 2019 - Sep 2021

KEY PROJECTS

IoT-Based Air Quality Index Meter System | *Final Year Project*

- Developed real-time air quality monitoring system using ESP32 with MQ-135, MQ-5, PM2.5 sensors, Firebase cloud integration, achieving 95%+ accuracy with web dashboard and mobile app

AI Medical Diagnosis System | *Machine Learning*

- Developed CNN-based deep learning model for automated medical image classification using transfer learning with ResNet/VGG architectures for disease detection from X-rays and CT scans

Financial Institution Network Design | *Computer Networks*

- Designed scalable three-tier network architecture using Cisco Packet Tracer supporting 500+ users with OSPF routing, VLAN segmentation, VPN connectivity, and 99.9% uptime

4-Bit Microprocessor Design | *Computer Architecture*

- Designed complete 4-bit CPU with custom ISA implementing ALU, control unit, and memory using Verilog/VHDL with fetch-decode-execute cycle validation

Industrial-Grade OMR Scanner | *Engineering Economics*

- Conducted cost-benefit analysis and feasibility study for industrial OMR scanner, achieving 40% cost reduction while maintaining accuracy through value engineering

Lung Segmentation from Chest X-rays | *Digital Image Processing*

- Developed automated lung segmentation algorithm using thresholding, morphological operations, edge detection, and region growing in MATLAB/Python

Digital Filter Design and Implementation | *Digital Signal Processing*

- Designed FIR and IIR digital filters using MATLAB with window methods and bilinear transformation, implementing real-time signal processing algorithms in C/Python

Speech Recognition System | *Signals and Systems*

- Implemented MFCC feature extraction with HMM/Neural Networks achieving 85-95% recognition accuracy on isolated word datasets using MATLAB

Automatic Vehicle Speed Controller | *Embedded Systems*

- Designed intelligent speed control system using Arduino with IR/ultrasonic sensors, implementing PID control and PWM for 90%+ accuracy in speed regulation

Online Library Management System | *Database Systems*

- Developed web-based library system with MySQL database featuring normalized schema (3NF), complex queries, achieving <200ms response time for 10K+ records

Prayer Alarm Mobile Application | *Mobile Development*

- Developed cross-platform prayer reminder app using Flutter with GPS-based Salah time calculations, Qibla compass, and offline support for 1000+ cities

Traffic Control System | *Digital Logic Design*

- Designed intelligent traffic light controller using FSM architecture with Verilog/VHDL on FPGA, implementing 4-way intersection control with <1ms state transition

Network Coding for Wireless Applications | *Wireless Networks*

- Conducted literature review analyzing network coding techniques (RLNC, PNC) for 5G/6G networks, IoT systems, and VANETs to improve throughput and reduce latency

Page Replacement Algorithm Implementation | *Operating Systems*

- Implemented and compared FIFO, LRU, Optimal, and Clock algorithms in C/C++, achieving 25-40% page fault reduction with LRU on real-world datasets

Grocery Management System | *Software Engineering*

- Developed full-stack POS system with inventory management, barcode scanning, and sales reporting using MVC architecture, reducing checkout time by 40%

Cricket Score Sheet System | *Data Structures*

- Built comprehensive cricket scoring application using linked lists, hash tables, and BST in C++/Java with $O(\log n)$ complexity for 1000+ match records

Galaxy Tab S8 CAD Modeling | *Computer Aided Design*

- Created detailed 3D model of Samsung Galaxy Tab S8 using SolidWorks with 25+ components, exploded views, and engineering drawings with GD&T

Hangman Game in C++ | *Object-Oriented Programming*

- Developed interactive word-guessing game applying OOP principles with encapsulation, inheritance, polymorphism, and STL containers for 500+ word database

Rain Detection System | *Digital Logic Design*

- Designed automated rain detection circuit using combinational and sequential logic with 555 timer, achieving optimal gate-level implementation with noise immunity

FM Long Range Transmission | *Engineering Workshop*

- Constructed FM radio transmitter operating at 88-108 MHz using LC oscillator and RF amplifier, achieving 1-3 km transmission range with clear audio quality

Temperature and Humidity Sensor with Arduino | *Microprocessor Interfacing*

- Interfaced DHT11/DHT22 sensor with Arduino using single-wire protocol, displaying real-time readings on LCD with data logging to SD card for HVAC applications

LU Decomposition with File Handling | *Programming Fundamentals*

- Implemented LU decomposition algorithm in C++ solving linear equations with file I/O operations for matrix factorization and solution vectors for 10x10+ systems

Serial Port Communication with Blink LEDs | *Embedded Systems*

- Implemented UART serial communication protocol on Arduino/STM32 for bidirectional data transfer controlling LED patterns with variable frequency via terminal commands

CERTIFICATIONS & TRAINING

- Smart Engineering with IoT and Cloud Integration - Pakistan Engineering Council
- Data Warehousing for Smart Infrastructure, Energy & Utilities - Pakistan Engineering Council
- Project Planning & Time Management - Pakistan Engineering Council
- Reliability Engineering of Complex Systems - Pakistan Engineering Council

- Construction OHS Code 2025 Framework & Implementation - Pakistan Engineering Council
- Robotics and AI in Healthcare Seminar - HITEC University
- CODE WAR Competition - HITEC University
- Speed Programming Certificate - NUST EME College
- Speed Coding in Robo Festa - HITEC University
- Internship Certificates - Machadev Engineering Pvt Ltd & Risetech